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CAPSTONE PROJECT

NOTEBOOK 24: MULTI-LABEL TARGET MATRIX AND SPLIT AUDIT

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Purpose:

This notebook prepares the full multi-label target space for

genre tagging using the genres_all metadata field.

The goal is to:

1. Load the large multi-label master table

2. Build full and candidate multi-label target matrices

3. Audit label frequencies by split

4. Separate core modelling labels from rare-tail labels

5. Save clean files for the next modelling phase

==============================

In [1]:

```
# ============================================================
# 1. IMPORT LIBRARIES
# ============================================================

import os
import ast
import numpy as np
import pandas as pd
from sklearn.preprocessing import MultiLabelBinarizer
```

In [2]:

```
# ============================================================
# 2. LOAD SAVED TABLES
# ============================================================

master_df = pd.read_csv("../data/processed/large_multilabel_master_table.csv")
genre_inventory = pd.read_csv("../data/processed/full_genre_inventory.csv")
candidate_genres = pd.read_csv("../data/processed/modelling_candidate_genres.csv")

print("Master table shape:", master_df.shape)
print("Genre inventory shape:", genre_inventory.shape)
print("Candidate genres shape:", candidate_genres.shape)

display(master_df.head())
display(genre_inventory.head())
display(candidate_genres.head())
```

```
Master table shape: (81574, 10)
Genre inventory shape: (163, 11)
Candidate genres shape: (150, 11)
```

	track_id	split	subset	genre_top	title	genres_ids	genres_all_ids	genres_all_names
0	20	training	large	NaN	Spiritual Level	76,103	17,10,76,103	Folk | Pop | Experimental Pop | Singer-Songwriter
1	26	training	large	NaN	Where is your Love?	76,103	17,10,76,103	Folk | Pop | Experimental Pop | Singer-Songwriter
2	30	training	large	NaN	Too Happy	76,103	17,10,76,103	Folk | Pop | Experimental Pop | Singer-Songwriter
3	46	training	large	NaN	Yosemite	76,103	17,10,76,103	Folk | Pop | Experimental Pop | Singer-Songwriter
4	48	training	large	NaN	Light of Light	76,103	17,10,76,103	Folk | Pop | Experimental Pop | Singer-Songwriter

	genre_id	genre_name	parent_id	parent_name	root_genre_id	root_genre_name	top_level_f
0	38	Experimental	NaN	NaN	38	Experimental	
1	15	Electronic	NaN	NaN	15	Electronic	
2	12	Rock	NaN	NaN	12	Rock	
3	1235	Instrumental	NaN	NaN	1235	Instrumental	1
4	10	Pop	NaN	NaN	10	Pop	

	genre_id	genre_name	parent_id	parent_name	root_genre_id	root_genre_name	top_level_f
0	38	Experimental	NaN	NaN	38	Experimental	
1	15	Electronic	NaN	NaN	15	Electronic	
2	12	Rock	NaN	NaN	12	Rock	
3	1235	Instrumental	NaN	NaN	1235	Instrumental	1
4	10	Pop	NaN	NaN	10	Pop	

In [3]:

```
# ============================================================
# 3. PARSE GENRE LIST COLUMNS
```

```
# ============================================================

def parse_csv_id_list(value):
    if pd.isna(value):
        return []
    text = str(value).strip()
    if text == "":
        return []
    return [int(x) for x in text.split(",") if str(x).strip() != ""]

master_df["genres_all_id_list"] =
master_df["genres_all_ids"].apply(parse_csv_id_list)
master_df["genres_id_list"] = master_df["genres_ids"].apply(parse_csv_id_list)

print("Parsed list columns added.")
display(master_df[["track_id", "genre_top", "genres_all_ids",
"genres_all_id_list"]].head(10))
```

```
Parsed list columns added.
```

	track_id	genre_top	genres_all_ids	genres_all_id_list
0	20	NaN	17,10,76,103	[17, 10, 76, 103]
1	26	NaN	17,10,76,103	[17, 10, 76, 103]
2	30	NaN	17,10,76,103	[17, 10, 76, 103]
3	46	NaN	17,10,76,103	[17, 10, 76, 103]
4	48	NaN	17,10,76,103	[17, 10, 76, 103]
5	135	Rock	58,12,45	[58, 12, 45]
6	137	Experimental	32,1,38	[32, 1, 38]
7	138	Experimental	32,1,38	[32, 1, 38]
8	142	Folk	17	[17]
9	144	Jazz	4	[4]

In [4]:

```
# ============================================================
# 4. DEFINE FULL AND CANDIDATE LABEL SPACES
# ============================================================

full_label_ids = sorted(genre_inventory["genre_id"].astype(int).tolist())
candidate_label_ids = sorted(candidate_genres["genre_id"].astype(int).tolist())

rare_tail_label_ids = sorted(set(full_label_ids) - set(candidate_label_ids))

print("Total full labels:", len(full_label_ids))
print("Total candidate labels:", len(candidate_label_ids))
print("Total rare-tail labels:", len(rare_tail_label_ids))
```

```python
print("\nFirst few rare-tail label IDs:", rare_tail_label_ids[:20])
```

```
Total full labels: 163
Total candidate labels: 150
Total rare-tail labels: 13

First few rare-tail label IDs: [173, 174, 175, 176, 178, 189, 374, 377, 465, 493,
808, 1032, 1060]
```

In [5]:

```python
# =============================================================
# 5. BUILD FULL 161-LABEL TARGET MATRIX
# =============================================================

mlb_full = MultiLabelBinarizer(classes=full_label_ids)
Y_full = mlb_full.fit_transform(master_df["genres_all_id_list"])

Y_full_df = pd.DataFrame(
    Y_full,
    columns=[f"genre_{gid}" for gid in mlb_full.classes_],
    index=master_df.index
)

print("Full target matrix shape:", Y_full_df.shape)
display(Y_full_df.head())
```

```
Full target matrix shape: (81574, 163)
```

	genre_1	genre_2	genre_3	genre_4	genre_5	genre_6	genre_7	genre_8	genre_9	genre_1
0	0	0	0	0	0	0	0	0	0	
1	0	0	0	0	0	0	0	0	0	
2	0	0	0	0	0	0	0	0	0	
3	0	0	0	0	0	0	0	0	0	
4	0	0	0	0	0	0	0	0	0	

5 rows × 163 columns

In [6]:

```python
# =============================================================
# 6. BUILD 150-LABEL CANDIDATE TARGET MATRIX
# =============================================================

mlb_candidate = MultiLabelBinarizer(classes=candidate_label_ids)
Y_candidate = mlb_candidate.fit_transform(master_df["genres_all_id_list"])

Y_candidate_df = pd.DataFrame(
```

```
    Y_candidate,
    columns=[f"genre_{gid}" for gid in mlb_candidate.classes_],
    index=master_df.index
)

print("Candidate target matrix shape:", Y_candidate_df.shape)
display(Y_candidate_df.head())
```

```
Candidate target matrix shape: (81574, 150)
```

```
c:\Users\jdevo\AppData\Local\Programs\Python\Python39\lib\site-packages\sklearn\preprocessing\_label.py:909: UserWarning: unknown class(es) [1032, 1060, 173, 174, 176, 189, 374, 377, 465, 493, 808] will be ignored
  warnings.warn(
```

	genre_1	genre_2	genre_3	genre_4	genre_5	genre_6	genre_7	genre_8	genre_9	genre_1
0	0	0	0	0	0	0	0	0	0	
1	0	0	0	0	0	0	0	0	0	
2	0	0	0	0	0	0	0	0	0	
3	0	0	0	0	0	0	0	0	0	
4	0	0	0	0	0	0	0	0	0	

5 rows × 150 columns

In [7]:

```
# ============================================================
# 7. BUILD LABEL NAME MAPS
# ============================================================

genre_name_map = dict(zip(
    genre_inventory["genre_id"].astype(int),
    genre_inventory["genre_name"].astype(str)
))

candidate_name_map = {gid: genre_name_map.get(gid, f"genre_{gid}") for gid in candidate_label_ids}
rare_tail_name_map = {gid: genre_name_map.get(gid, f"genre_{gid}") for gid in rare_tail_label_ids}

print("Example candidate labels:")
for gid in candidate_label_ids[:10]:
    print(gid, "->", candidate_name_map[gid])

print("\nRare-tail labels:")
for gid in rare_tail_label_ids[:20]:
    print(gid, "->", rare_tail_name_map[gid])
```

```
Example candidate labels:
1 -> Avant-Garde
2 -> International
3 -> Blues
4 -> Jazz
5 -> Classical
6 -> Novelty
7 -> Comedy
8 -> Old-Time / Historic
9 -> Country
10 -> Pop

Rare-tail labels:
173 -> N. Indian Traditional
174 -> South Indian Traditional
175 -> Bollywood
176 -> Pacific
178 -> Be-Bop
189 -> Talk Radio
374 -> Banter
377 -> Deep Funk
465 -> Musical Theater
493 -> Western Swing
808 -> Salsa
1032 -> Turkish
1060 -> Tango
```

In [8]:

```python
# ============================================================
# 8. AUDIT LABEL FREQUENCIES FOR FULL LABEL SPACE
# ============================================================

full_label_counts = Y_full_df.sum(axis=0).sort_values(ascending=False)

full_label_summary = pd.DataFrame({
    "genre_id": [int(col.replace("genre_", "")) for col in full_label_counts.index],
    "genre_name": [genre_name_map[int(col.replace("genre_", ""))] for col in full_label_counts.index],
    "total_count": full_label_counts.values
})

print("Top full-label counts:")
display(full_label_summary.head(30))
```

```
Top full-label counts:
```

	genre_id	genre_name	total_count
0	38	Experimental	35903
1	15	Electronic	28099
2	12	Rock	25820
3	1235	Instrumental	13588
4	10	Pop	12659
5	17	Folk	11187
6	1	Avant-Garde	8134
7	107	Ambient	6799
8	32	Noise	6781
9	76	Experimental Pop	6632
10	21	Hip-Hop	6188
11	25	Punk	5942
12	41	Electroacoustic	5793
13	27	Lo-Fi	5492
14	18	Soundtrack	5092
15	42	Ambient Electronic	4719
16	2	International	4253
17	66	Indie-Rock	4139
18	250	Improv	4031
19	4	Jazz	3742
20	103	Singer-Songwriter	3588
21	5	Classical	3487
22	30	Field Recordings	3037
23	247	Musique Concrete	2830
24	236	IDM	2484
25	47	Drone	2340
26	183	Glitch	2260
27	85	Garage	2131
28	33	Psych-Folk	2123
29	70	Industrial	2058

In [9]:

```
# ============================================================
# 9. AUDIT LABEL FREQUENCIES FOR CANDIDATE LABEL SPACE
# ============================================================

candidate_label_counts = Y_candidate_df.sum(axis=0).sort_values(ascending=False)

candidate_label_summary = pd.DataFrame({
    "genre_id": [int(col.replace("genre_", "")) for col in
candidate_label_counts.index],
    "genre_name": [genre_name_map[int(col.replace("genre_", ""))] for col in
candidate_label_counts.index],
    "total_count": candidate_label_counts.values
})

print("Top candidate-label counts:")
display(candidate_label_summary.head(30))
```

```
Top candidate-label counts:
```

	genre_id	genre_name	total_count
0	38	Experimental	35903
1	15	Electronic	28099
2	12	Rock	25820
3	1235	Instrumental	13588
4	10	Pop	12659
5	17	Folk	11187
6	1	Avant-Garde	8134
7	107	Ambient	6799
8	32	Noise	6781
9	76	Experimental Pop	6632
10	21	Hip-Hop	6188
11	25	Punk	5942
12	41	Electroacoustic	5793
13	27	Lo-Fi	5492
14	18	Soundtrack	5092
15	42	Ambient Electronic	4719
16	2	International	4253
17	66	Indie-Rock	4139
18	250	Improv	4031
19	4	Jazz	3742
20	103	Singer-Songwriter	3588
21	5	Classical	3487
22	30	Field Recordings	3037
23	247	Musique Concrete	2830
24	236	IDM	2484
25	47	Drone	2340
26	183	Glitch	2260
27	85	Garage	2131
28	33	Psych-Folk	2123
29	70	Industrial	2058

In [10]:

```
# ============================================================
# 10. SPLIT-LEVEL LABEL AUDIT FOR CANDIDATE LABELS
# ============================================================

split_series = master_df["split"].astype(str)

candidate_split_rows = []

for split_name in ["training", "validation", "test"]:
    mask = (split_series == split_name).values
    split_counts = Y_candidate_df.loc[mask].sum(axis=0)

    temp_df = pd.DataFrame({
        "genre_id": [int(col.replace("genre_", "")) for col in split_counts.index],
        "genre_name": [genre_name_map[int(col.replace("genre_", ""))] for col in
split_counts.index],
        "split": split_name,
        "count": split_counts.values
    })

    candidate_split_rows.append(temp_df)

candidate_split_summary = pd.concat(candidate_split_rows, ignore_index=True)

print("Candidate split summary shape:", candidate_split_summary.shape)
display(candidate_split_summary.head(30))
```

```
Candidate split summary shape: (450, 4)
```

	genre_id	genre_name	split	count
0	1	Avant-Garde	training	6405
1	2	International	training	3311
2	3	Blues	training	1343
3	4	Jazz	training	2747
4	5	Classical	training	2784
5	6	Novelty	training	612
6	7	Comedy	training	160
7	8	Old-Time / Historic	training	274
8	9	Country	training	1443
9	10	Pop	training	10056
10	11	Disco	training	245
11	12	Rock	training	20138
12	13	Easy Listening	training	520
13	14	Soul-RnB	training	1105
14	15	Electronic	training	22263
15	16	Sound Effects	training	168
16	17	Folk	training	8745
17	18	Soundtrack	training	4043
18	19	Funk	training	566
19	20	Spoken	training	1245
20	21	Hip-Hop	training	4665
21	22	Audio Collage	training	500
22	25	Punk	training	4782
23	26	Post-Rock	training	1114
24	27	Lo-Fi	training	4044
25	30	Field Recordings	training	2450
26	31	Metal	training	707
27	32	Noise	training	5158
28	33	Psych-Folk	training	1681
29	36	Krautrock	training	484

In [11]:

```
# ============================================================
# 11. LABEL COVERAGE SUMMARY TABLE
# ============================================================

coverage_summary = pd.DataFrame({
    "Label Space": ["Full genres_all", "Candidate modelling labels", "Rare-tail
labels"],
    "Number of Labels": [len(full_label_ids), len(candidate_label_ids),
len(rare_tail_label_ids)],
    "Description": [
        "All genres present in metadata",
        "Genres with at least 25 occurrences in the large audio subset",
        "Genres below the candidate threshold"
    ]
})

print("Label coverage summary:")
display(coverage_summary)
```

```
Label coverage summary:
```

	Label Space	Number of Labels	Description
0	Full genres_all	163	All genres present in metadata
1	Candidate modelling labels	150	Genres with at least 25 occurrences in the lar...
2	Rare-tail labels	13	Genres below the candidate threshold

In [12]:

```
# ============================================================
# 12. BUILD MODELLING MASTER TABLES
# ============================================================

multilabel_full_master = pd.concat(
    [master_df[["track_id", "split", "subset", "genre_top", "title", "audio_path",
"audio_exists"]], Y_full_df],
    axis=1
)

multilabel_candidate_master = pd.concat(
    [master_df[["track_id", "split", "subset", "genre_top", "title", "audio_path",
"audio_exists"]], Y_candidate_df],
    axis=1
)

print("Full multi-label modelling table shape:", multilabel_full_master.shape)
print("Candidate multi-label modelling table shape:",
multilabel_candidate_master.shape)
```

```
Full multi-label modelling table shape: (81574, 170)
Candidate multi-label modelling table shape: (81574, 157)
```

In [13]:

```python
# ============================================================
# 13. SAVE OUTPUT FILES
# ============================================================

os.makedirs("../data/processed", exist_ok=True)

full_label_summary.to_csv(
    "../data/processed/full_label_summary.csv",
    index=False
)

candidate_label_summary.to_csv(
    "../data/processed/candidate_label_summary.csv",
    index=False
)

candidate_split_summary.to_csv(
    "../data/processed/candidate_split_summary.csv",
    index=False
)

coverage_summary.to_csv(
    "../data/processed/multilabel_label_coverage_summary.csv",
    index=False
)

multilabel_full_master.to_csv(
    "../data/processed/multilabel_full_master_table.csv",
    index=False
)

multilabel_candidate_master.to_csv(
    "../data/processed/multilabel_candidate_master_table.csv",
    index=False
)

print("Saved:")
print("- ../data/processed/full_label_summary.csv")
print("- ../data/processed/candidate_label_summary.csv")
print("- ../data/processed/candidate_split_summary.csv")
print("- ../data/processed/multilabel_label_coverage_summary.csv")
print("- ../data/processed/multilabel_full_master_table.csv")
print("- ../data/processed/multilabel_candidate_master_table.csv")
```

```
Saved:
- ../data/processed/full_label_summary.csv
- ../data/processed/candidate_label_summary.csv
- ../data/processed/candidate_split_summary.csv
- ../data/processed/multilabel_label_coverage_summary.csv
- ../data/processed/multilabel_full_master_table.csv
- ../data/processed/multilabel_candidate_master_table.csv
```

In [14]:

```python
# ============================================================
# 14. INTERPRETATION NOTES
# ============================================================

print("1. The full metadata target space contains 161 unique genres.")
print("2. A candidate modelling space of 150 labels was created using a minimum-count threshold.")
print("3. The remaining rare-tail labels are still preserved for inventory and future hierarchical handling.")
print("4. The saved multi-label master tables are the basis for full metadata genre identification.")
print("5. The next modelling phase should begin with structured multi-label baselines on the candidate label space.")
```

```
1. The full metadata target space contains 161 unique genres.
2. A candidate modelling space of 150 labels was created using a minimum-count
threshold.
3. The remaining rare-tail labels are still preserved for inventory and future
hierarchical handling.
4. The saved multi-label master tables are the basis for full metadata genre
identification.
5. The next modelling phase should begin with structured multi-label baselines on the
candidate label space.
```